# Hi3D v1.0: The First Production-Validated Open 3D Foundation Model

Code (GitHub) | Model weights (Hugging Face)

**Abstract**

While recent image-to-3D models excel at mimicking the appearance of an input view, practical 3D creation demands far more than reference-view similarity. Downstream applications—including physics simulation, animation, interactive editing, and 3D printing—require full-geometry fidelity, robust thin-structure preservation, and multi-view geometric plausibility. In this work, we introduce Hi3D v1.0, a state-of-the-art open-source 3D foundation model directly validated through large-scale commercial and industrial production. Built upon the sparse-convolutional framework of Sparc3D [1], Hi3D v1.0 scales generative pre-training to approximately two million curated, high-quality 3D assets and builds on the $1536^3$ image-to-3D generation capability pioneered by our earlier Sparc3D system. By integrating high-resolution geometric supervision within a production-proven architecture, Hi3D v1.0 bridges the critical divide between 2D perceptual appeal and robust, watertight 3D utility. Comprehensive quantitative benchmarks and a 562-judgment human preference study demonstrate that Hi3D v1.0 achieves an aggregate preference score of 85.9%, establishing a new state of the art among open-source image-to-3D systems and demonstrating the efficacy of industry-scale data and resolution scaling for usable 3D asset generation.

## 1 Introduction

Image-conditioned 3D generation is increasingly good at reproducing what is visible in a reference image. Higher-capacity representations and stronger image conditioning have made the rendered result from the input camera a useful and demanding measure of progress. Yet this view captures only part of the output. A generated asset is ultimately rotated, inspected, edited, placed in a scene, passed to geometry-processing software, or prepared for fabrication. Large portions of its sides, back, bottom, and occluded structures may never be observed by an input-view metric.

This creates a practical distinction between *input-view fidelity* and *holistic 3D quality*. The former asks whether the generated asset resembles the conditioning image from the conditioning camera. The latter asks whether the result remains a convincing and usable three-dimensional object: whether its unseen regions are geometrically plausible, whether thin and disconnected structures survive, whether local detail is organized as geometry rather than surface noise, and whether the mesh satisfies the requirements of downstream tools. These criteria should complement input-view evaluation rather than replace it.

**A 3D asset is not a rendered image.**
**Front-view fidelity is not 3D quality.**

We revisit this distinction through Hi3D v1.0, the open release of our production image-to-3D system. Its technical basis is Sparc3D, which introduced a high-resolution sparse representation and sparse-convolutional autoencoding pipeline for detailed geometry[1]. The main changes reported here are deliberately straightforward: the production checkpoint was trained at a scale of approximately two million 3D assets, training was extended with data for the $1536^3$ operating level, and the resulting system was exercised in a real user-facing deployment before release. We do not claim a new generative objective, decoder, or surface representation in this report.

Single-image 3D generation is inherently underconstrained: the input provides direct evidence for only part of the surface, while the remaining geometry must be inferred from learned three-dimensional priors. This makes the scale and diversity of the training corpus particularly important,

as broader geometric coverage can strengthen the model's ability to reconstruct unseen regions and long-tail structures. Motivated by this observation, we expand the training corpus and include a subset trained at $1536^3$ resolution to improve high-resolution geometric reconstruction.

The central question of this report is whether a scaled training corpus and additional high-resolution training data can produce a stronger general-purpose checkpoint within the existing Sparc3D framework. We evaluate the resulting system beyond the conditioning view, with particular attention to unseen geometry, thin structures, local geometric detail, and consistency across multiple viewpoints. This evaluation includes matched qualitative comparisons and a 562-judgment user study designed to assess properties that are not fully captured by input-view similarity alone.

Our contributions are therefore system- and release-oriented:

- **An open, production-validated release.** We release Hi3D v1.0, a Sparc3D-based image-to-3D system whose checkpoint has been exercised in a real production workflow before open publication.
- **Training at the two-million-asset scale.** We expand the controlled Sparc3D training setting to approximately two million curated 3D assets and include training data for a $1536^3$ high-resolution operating level.
- **Evaluation beyond the input view.** We report image-conditioned and geometric evaluations, a user study with 562 judgments from 13 participants, and matched front, back, bottom, and local-detail comparisons.

# 2 Hi3D v1.0 Overview

## 2.1 A Production-Scale Sparc3D System

Hi3D v1.0 retains the core geometry pipeline of Sparc3D. At a high level, an upstream image-to-coarse-mesh stage provides the global structure. Sparc3D refinement then uses the reference image and coarse mesh together: image features condition a sparse flow model on geometry-derived coordinates, a self-pruning sparse VAE decoder expands those features around predicted surface regions, and a sparse isosurface extractor produces the refined mesh. Readers seeking the representation, training objective, decoder, and extraction details should refer to the original Sparc3D report[1].

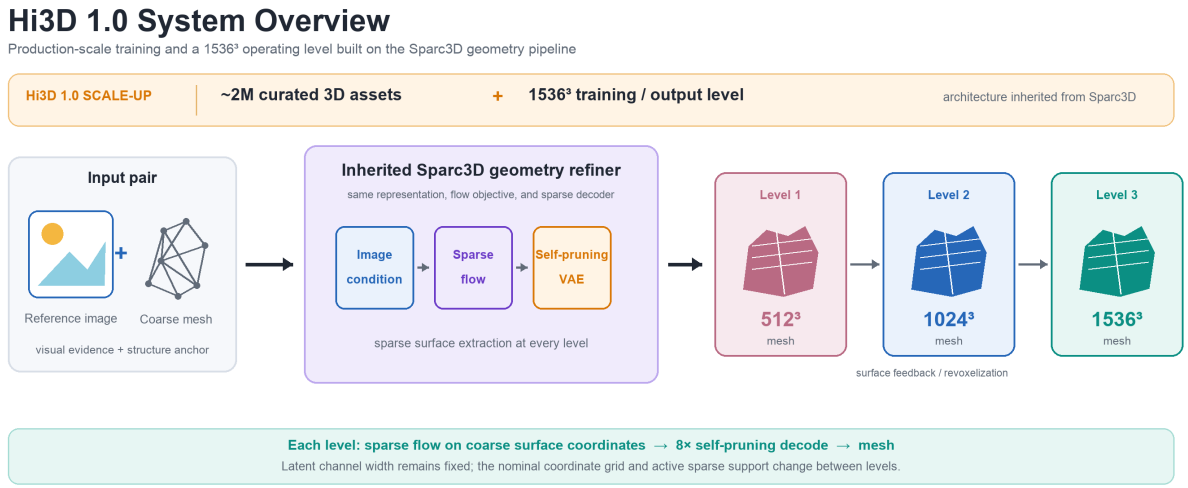



**Figure 1:** Compact overview of Hi3D v1.0. The release retains the Sparc3D geometry refiner and scales its training data and resolution ladder. A reference image and coarse mesh condition sparse refinement; each level evolves latent features on coarse surface coordinates, applies an $8\times$ self-pruning decoder, extracts a mesh, and revoxelizes that mesh before the next level.

This division of labor is important. The flow transformer does not directly store a dense $512^3$, $1024^3$, or $1536^3$ latent grid. At the first level, latent features live on a sparse coordinate set defined on a nominal $64^3$ structural grid, and the decoder applies three learned $2\times$ expansions to extract a $512^3$-level surface. The resulting mesh can already be used as an output. To continue, it is revoxelized at the next structural level, producing new sparse coordinates for another flow trajectory and another $8\times$ decoder path. The latent feature width remains fixed, while the nominal coordinate grid and active support change between levels.

### 2.2 From $512^3$ to $1536^3$

The three operating levels use nominal structural grids of $64^3$, $128^3$, and $192^3$. With the fixed $8\times$ decoder scale, these yield nominal surface extraction grids of $512^3$, $1024^3$, and $1536^3$, respectively. The latent feature width remains fixed across levels, while the coordinate-grid resolution and active support change. All three levels use sparse grids; no dense volumetric grid is materialized.

Compared with the 500K-asset training setting reported for Sparc3D [1], Hi3D v1.0 uses approximately two million curated assets and adds a $1536^3$ training and output level.

The inherited Sparcubes preprocessing converts training geometry into a normalized high-resolution surface representation suitable for sparse learning [1].

## 3 Evaluation

We evaluate Hi3D v1.0 from both conventional image-conditioned fidelity and holistic 3D asset quality. In addition to ULIP-2 [2] and Uni3D [3], which primarily measure semantic consistency with the conditioning image, we report Multi-View Fidelity (**MVF**), Detail Richness (**DR**), and Slicing Success Rate (**SSR**) to evaluate the producibility and usability of the generated 3D asset.

For *Multi-View Fidelity*, each generated asset is rendered from six canonical viewpoints (front, back, left, right, top, and bottom). We assess (i) geometry consistency, which measures whether the object maintains coherent shape and structure across viewpoints, and (ii) an artifact penalty for structural failures, including broken or missing parts, floating fragments, duplicated geometry, and view-dependent hallucinations. Both terms are normalized to $[0, 1]$, and the per-asset score is defined as

$$S_{\mathrm{MV}} = \frac{1}{2}(S_{\mathrm{cons}} + 1 - S_{\mathrm{art}}), \tag{1}$$

where $S_{\mathrm{cons}}$ denotes geometric consistency and $S_{\mathrm{art}}$ denotes artifact severity. We report Multi-View Fidelity as the mean $S_{\mathrm{MV}}$ over all evaluation assets.

*Detail Richness* evaluates geometric detail independently of appearance. We use close-up clay renderings together with surface-normal maps to suppress the influence of texture and material, and assess four aspects of the generated geometry: fine-scale surface relief, structural layering, preservation of small components, and geometric detail outside the conditioning view. The per-asset *Detail Richness* score is the mean of four normalized component scores:

$$S_{\mathrm{detail}} = \frac{1}{4}(S_{\mathrm{relief}} + S_{\mathrm{layer}} + S_{\mathrm{small}} + S_{\mathrm{unseen}}). \tag{2}$$

We report Detail Richness as the mean score across all evaluation assets.

We evaluate slicing success for each generated mesh using the PrusaSlicer-family CLI in Bambu Studio. Each asset is uniformly scaled so the longest axis-aligned bounding-box edge is 100 mm, and placed with $Z_{\min} = 0$ and the $XY$ footprint centered on the bed; no explicit mesh repair (hole filling, manifold forcing, or non-manifold cleanup) is applied before slicing. Real FDM slicing uses fixed profiles—Bambu Lab A1 0.4 mm nozzle, process `0.20mm Standard @BBL A1`, and filament `Bambu PLA Basic @BBL A1`—with `--orient 0`, `--arrange 1`, and `--slice 0`. A trial counts as a *slice success* if the CLI produces a non-trivial plate G-code artifact larger than 200 bytes; it counts as a *failure* if `number_of_parts` $> 400$, the process exceeds an adaptive wall-clock budget (base 90 s), or Bambu returns a slicing error. For each method, the slicing success rate is the fraction of all test assets with `slice_success=True`.

**Table 1:** Quantitative comparison across geometric resolution, multi-view fidelity (MVF), detail richness (DR), and slicing success rate (SSR).

| | Reso. | ULIP-2 | Uni3D | MVF | DR | SSR |
|---|---|---|---|---|---|---|
| Hi3DGen [4] | 256 | 44.08 | 38.99 | 0.71 | 0.633 | 97% |
| TRELLIS [5] | | 45.93 | 39.25 | 0.67 | 0.396 | 97% |
| Hunyuan3D-2.1 [6] | 384 | 45.16 | 40.83 | 0.74 | 0.431 | 97% |
| TripoSG [7] | 512 | 45.16 | 40.13 | 0.72 | 0.366 | **100%** |
| Direct3D-S2 [8] | 1024 | 45.83 | 40.51 | 0.66 | 0.627 | 80% |
| UltraShape-1.0 [9] | | 46.81 | 41.59 | 0.74 | 0.632 | 97% |
| TRELLIS.2 [10] | | **49.32** | **44.18** | 0.74 | 0.677 | 17% |
| Pixal3D [11] | **1536** | 49.07 | 43.57 | 0.7 | 0.732 | 7% |
| **Hi3D v1.0** | | 48.07 | 42.14 | **0.75** | **0.841** | 90% |

**Table 2:** User preference score. The preference score assigns 1.0 to a Hi3D v1.0 win and 0.5 to a tie.

| Baseline | Judgments | Hi3D v1.0 better | Approximately equal | Baseline better | Preference score |
|---|---|---|---|---|---|
| Hi3DGen | 65 | 62 (95.4%) | 2 (3.1%) | 1 (1.5%) | 96.9% |
| TRELLIS | 68 | 65 (95.6%) | 1 (1.5%) | 2 (2.9%) | 96.3% |
| TripoSG | 76 | 73 (96.1%) | 0 (0.0%) | 3 (3.9%) | 96.1% |
| Hunyuan3D-2.1 | 72 | 65 (90.3%) | 2 (2.8%) | 5 (6.9%) | 91.7% |
| Direct3D-S2 | 70 | 62 (88.6%) | 2 (2.9%) | 6 (8.6%) | 90.0% |
| UltraShape-1.0 | 68 | 53 (77.9%) | 5 (7.4%) | 10 (14.7%) | 81.6% |
| TRELLIS.2 | 71 | 52 (73.2%) | 8 (11.3%) | 11 (15.5%) | 78.9% |
| Pixal3D | 72 | 37 (51.4%) | 8 (11.1%) | 27 (37.5%) | 56.9% |
| Overall | 562 | 469 (83.5%) | 28 (5.0%) | 65 (11.6%) | 85.9% |

### 3.1 Quantitative Comparison

As shown in Table 1, Hi3D v1.0 achieves advanced geometric quality while maintaining strong printability. It operates at the highest evaluated geometric resolution of **1536**, while achieving the best multi-view fidelity (**MVF**) of **0.75** and detail richness (**DR**) of **0.841** among all compared methods. Notably, these gains in geometric expressiveness do not come at the severe cost of manufacturability observed in other high-resolution methods: Hi3D v1.0 maintains a **90%** slicing success rate (**SSR**), compared with only **7%** for Pixal3D at the same resolution and **17%** for TRELLIS.2. These results demonstrate that Hi3D v1.0 effectively combines high-resolution geometry, strong multi-view consistency, and rich geometric details with robust compatibility for downstream 3D printing.

### 3.2 User Preference Study

We conducted a pairwise A/B user study to evaluate perceived geometry quality. For each comparison, participants were shown both the mesh renderings and corresponding normal maps produced by Hi3D v1.0 and one competing method. They were asked to select the result with better overall geometry quality or indicate a tie. Thirteen participants, all with professional experience in 3D algorithms, contributed 562 valid judgments.

Across all comparisons, Hi3D v1.0 was preferred in 469 judgments (83.5%), judged approximately equal in 28 (5.0%), and disfavored in 65 (11.6%). Assigning half credit to ties gives an aggregate preference score of 85.9%; after excluding ties, the win rate is 87.8%.

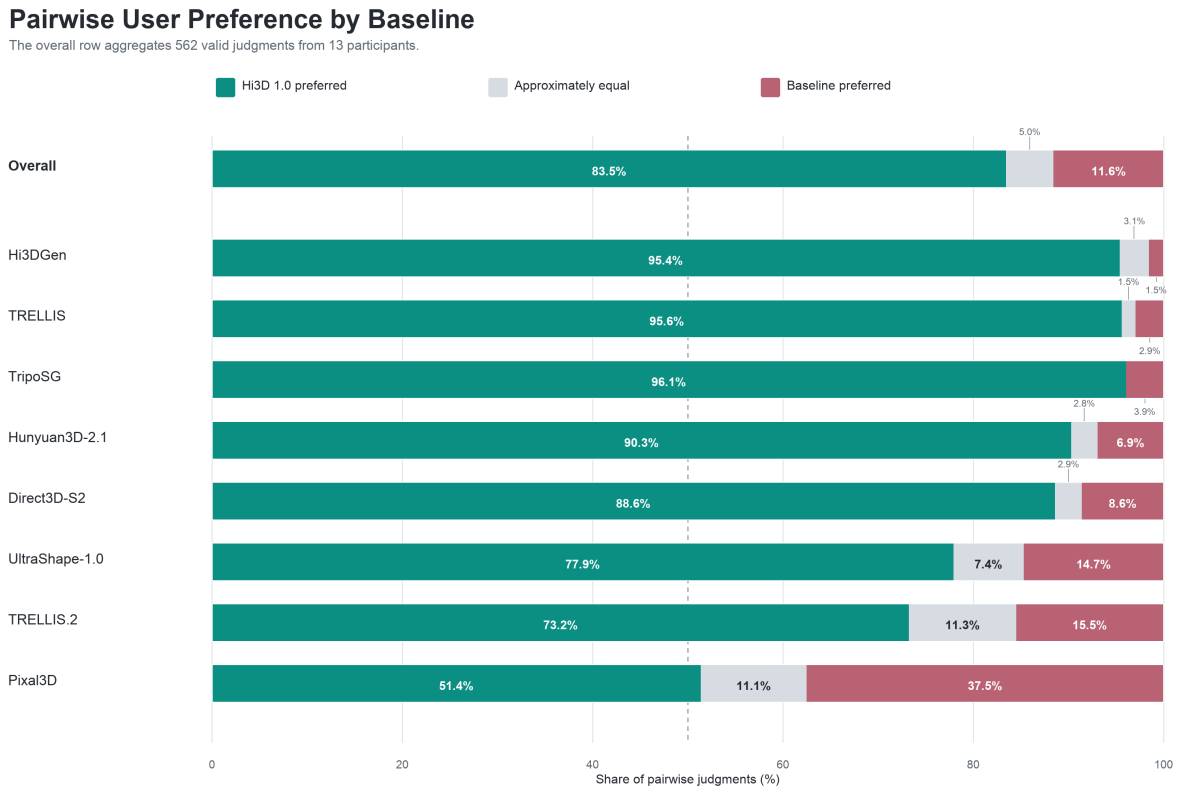



**Figure 2:** Pairwise A/B user-preference results comparing Hi3D v1.0 with eight image-to-3D baselines. Participants with professional experience in 3D algorithms evaluated both mesh renderings and corresponding normal maps. The overall result aggregates 562 valid judgments.

### 3.3 Qualitative Comparison

Figures 3 and 4 compare full-object mesh renderings and surface-normal maps across the evaluated methods. Hi3D v1.0 consistently preserves the global silhouette and connectivity of major components while retaining finer local geometry. For mechanically structured objects, including the armored vehicle, mechanical creature, and robot, Hi3D v1.0 preserves major components and their connections, as well as local features such as recessed panels and attachment boundaries. Several competing methods either smooth away these structures or introduce fragmented, inflated, or flattened geometry. The winged armored figure also illustrates the preservation of thin protrusions and layered structures while maintaining the overall shape.

The close-up comparisons in Figure 5 further reveal differences in local surface quality. The bird and dragon examples illustrate the preservation of local features such as feathers, horns, and teeth. Hi3D v1.0 reconstructs repeated feather structures, architectural grooves, layered relief, circular recesses, and scale-like surface patterns with clearer separation between adjacent components. The corresponding normal maps exhibit structured variations aligned with visible geometric boundaries, rather than unorganized high-frequency noise. On smooth manufactured surfaces, Hi3D v1.0 retains seams, openings, and panel transitions while avoiding the excessive roughness observed in some competing results. Geometry outside the conditioning view is generally less detailed than image-visible geometry but remains structurally coherent.

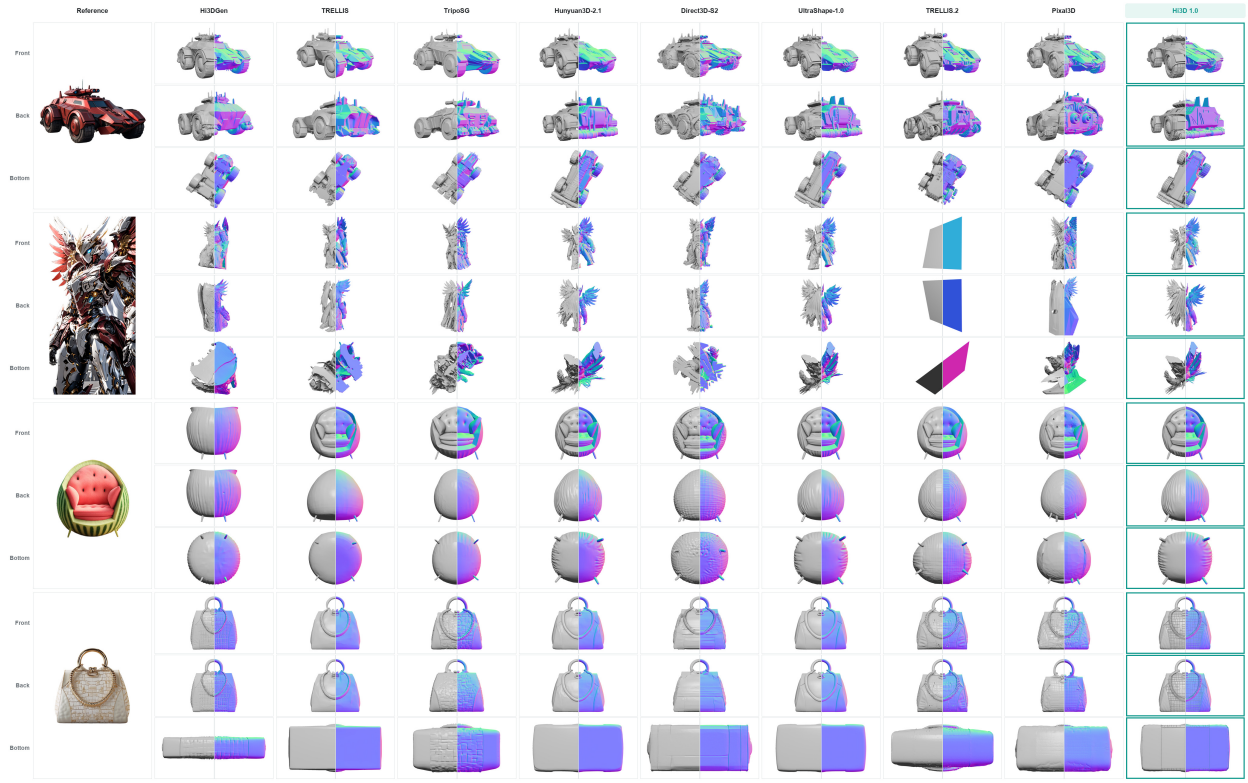



**Figure 3:** Global qualitative comparison on four objects. Rows show front, back, and bottom views under matched cameras, normalization, scale, and crop. These examples visualize the distinction between input-view fidelity and complete-object behavior. Since the single input image does not reveal the true backside, the hidden views provide evidence of plausibility and structural consistency, not exact unseen-surface reconstruction.

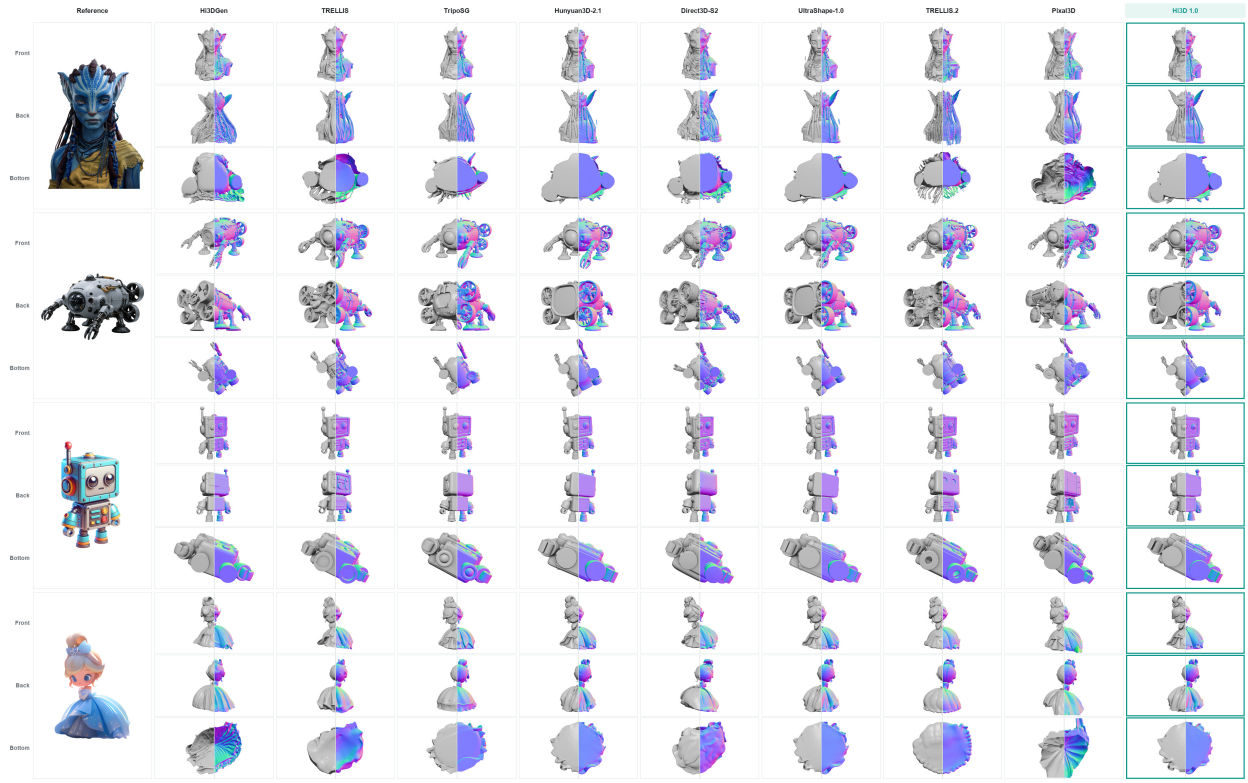



**Figure 4:** Additional matched front, back, and bottom comparisons across organic, mechanical, and articulated objects. Hi3D v1.0 is highlighted in teal.

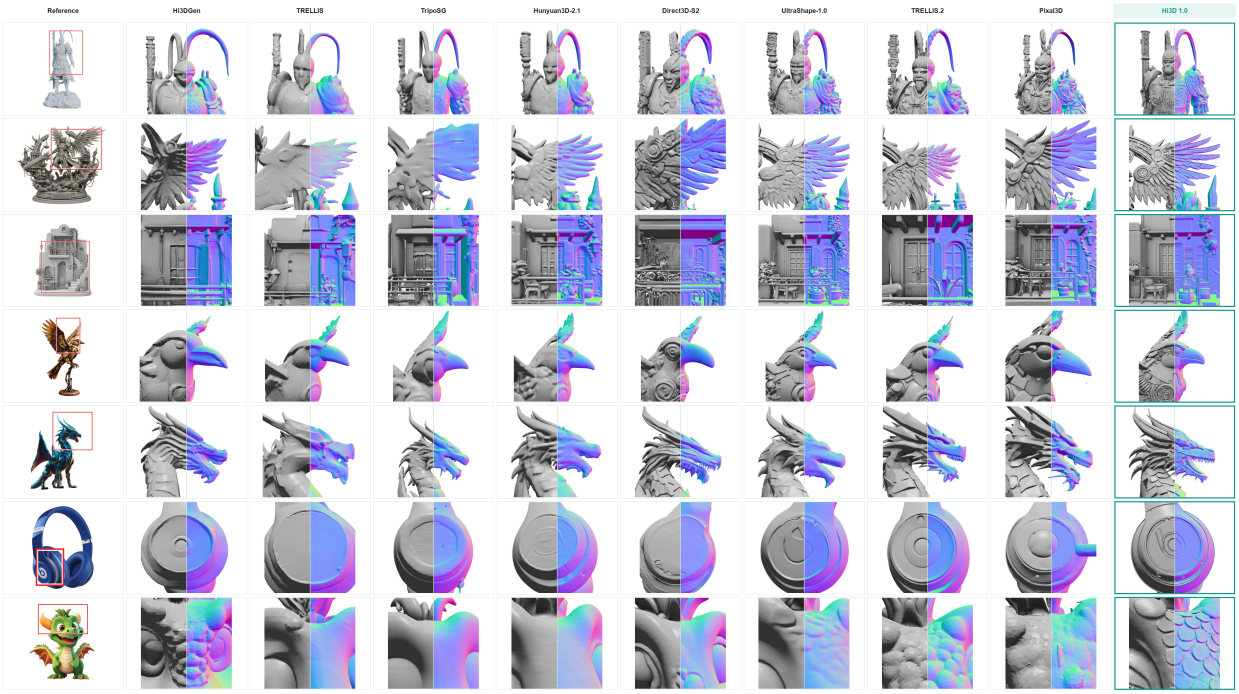



**Figure 5:** Local-detail comparison on seven diagnostic regions. Red boxes indicate the corresponding reference-image regions. Shaded geometry and normal renderings are shown together to distinguish coherent relief from irregular high-frequency surface noise. The selected regions include thin layers, armor relief, architectural moldings, curved boundaries, scales, teeth, and concentric grooves.

## 4 Discussion and Limitations

Our results reinforce a central observation: front-view fidelity alone is not a sufficient measure of image-to-3D quality. A practically useful 3D asset must remain geometrically coherent under novel viewpoints, preserve thin and high-frequency structures, and satisfy basic geometric requirements for downstream use. Hi3D v1.0 addresses these requirements through production-scale training, high-resolution sparse refinement up to $1536^3$, and extensive deployment in real-world creation workflows. Rather than introducing a new 3D representation, it demonstrates how the Sparc3D framework can be scaled into a robust and practically usable geometry-generation system.

Hi3D v1.0 should also be understood as the first open release in a continuously evolving production model family, rather than the endpoint of our system development. Since the development of Hi3D v1.0, our production system has progressed to Hi3D v3.0, which increases the nominal refinement resolution to $2048^3$ and further improves the recovery of fine geometric structures. We release Hi3D v1.0 as a stable and extensively validated foundation for the open-source community. A controlled comparison with Hi3D 3.0 is outside the scope of this report, and nominal resolution alone should not be interpreted as a complete measure of 3D quality.

Several limitations remain. Because the system operates from a single input image, geometry in fully occluded regions represents a plausible completion rather than a uniquely recoverable reconstruction. Hi3D v1.0 also depends on an upstream coarse mesh, and severe structural errors in this initialization may persist after refinement. The current release focuses on geometry and does not generate textures, materials, or vertex colors.

Finally, geometric validity is application dependent. Watertightness and successful slicing are useful indicators for fabrication-oriented workflows, but they do not guarantee physical manufacturability. Similarly, image-based preference captures perceptual quality but does not replace multi-view geometric evaluation. Future evaluation should therefore combine visual preference, held-out multi-view consistency, geometric validity, and application-specific success rates.

## 5 Conclusion

We release Hi3D v1.0, a production-deployed image-to-3D model based on Sparc3D, trained on approximately two million curated assets and supporting sparse geometry refinement up to $1536^3$. On our evaluation set, it obtains the highest MVF and DR scores among the compared methods, with a 90% slicing success rate. A 562-judgment user study yields an aggregate preference score of 85.9% with half credit for ties. The release provides an open checkpoint for generating and evaluating 3D geometry beyond the conditioning view.

## 6 Contributions and Acknowledgments

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